

Ahmad Rahimi

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[Homepage](#)

SUMMARY PhD candidate at EPFL specializing in efficient video generation, driving world models, and 3D/4D scene reconstruction. Published at CVPR 2022, 2025, and 2026 and ICML 2026, with industry research experience on Wayve’s GAIA team. Developed controllable video-diffusion models and dynamic Gaussian-splat reconstruction systems. Available from March 2027.

EDUCATION ♦ **Ph.D. Degree in Computer Science** Sept. 2022 – Expected Feb. 2027
Department of Computer Science Supervisor: Alexandre Alahi
École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

WORK EXPERIENCE ♦ **Applied Science Intern at Wayve** Nov. 2025 - May 2026
Worked with the GAIA team on driving world models and data-centric evaluation.

- ♦ Built a pipeline for generating 360° panoramic videos from recorded driving scenarios and a Meta Quest-based VR review application placing reviewers inside the vehicle environment. Improved labeling quality and increased intervention-retrieval recall by more than 3x.
- ♦ Designed feed-forward and recursive-refinement architectures for explicit 4D driving-scene reconstruction and generation with dynamic 3D Gaussian splats, using rendered intermediate predictions and conditioning images to iteratively improve scene estimates.

SELECTED PUBLICATIONS ♦ **“MAD: Motion Appearance Decoupling for efficient Driving World Models”**, Ahmad Rahimi*, Valentin Gerard*, Eloi Zablocki, Matthieu Cord, and Alexandre Alahi
Introduces a two-stage driving world model that decouples motion forecasting from appearance synthesis, enabling efficient video generation with only 140 GPU hours of training, two orders of magnitude less than prior work. (CVPR 2026), [project page](#)

♦ **“GEM: A Generalizable Ego-vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control”**, Mariam Hassan*, Sebastian Stapf*, Ahmad Rahimi*, and 17 more authors.
Develops a controllable ego-vision world model for long-horizon driving-video generation using multimodal controls over ego-motion, object dynamics, and scene composition, built on a Stable Video Diffusion backbone. (CVPR 2025), [link to arxiv](#)

♦ **From Generation to Generalization: Emergent Few-Shot Learning in Video Diffusion Models** Pablo Acuaviva, Aram Davtyan, Mariam Hassan, Sebastian Stapf, Ahmad Rahimi, Alexandre Alahi, and Paolo Favaro.
Demonstrates that pretrained video diffusion models can be adapted from only a few examples to diverse visual tasks, ranging from segmentation and keypoint estimation to maze solving and ARC-style reasoning, using lightweight LoRA fine-tuning. Comparison with LLMs suggests better sample efficiency. (ICML 2026, ARC Prize 2025 Honorable Mention), [link to arxiv](#)

♦ **“PACT: Preserving Accurate Continuous Trajectories via Flow-Map Distillation”**, Ahmad Rahimi and Alexandre Alahi.
A lightweight distillation approach for large-scale generative models, unlocking real-time video generation on a single GPU. Unlike existing methods that suffer from mode collapse, PACT strictly preserves the teacher’s seed-level behaviors and diversity, critical for world models that must capture varied yet controllable real-world dynamics. under review at (ICLR 2027)

OTHER PUBLICATIONS	◇ “Sim-to-Real Causal Transfer: A Metric Learning Approach to Causally-Aware Interaction Representations”, Ahmad Rahimi*, Po-Chien Luan*, Yuejiang Liu*, Frano Rajič, and Alexandre Alahi. (CVPR 2025), link to arxiv ◇ “Vehicle trajectory prediction works, but not everywhere”, Mohammadhossein Bahari*, Saeed Saadatnejad*, Ahmad Rahimi, Mohammad Shahverdi Kordori, Seyed-Mohsen Moosavi-Dezfooli, and Alexandre Alahi. (CVPR 2022), link to arxiv ◇ “A Multi-Loss Strategy for Vehicle Trajectory Prediction: Combining Off-Road, Diversity, and Directional Consistency Losses”, Ahmad Rahimi and Alexandre Alahi. link to arxiv
HONORS AND AWARDS	◇ Third Place in LLM Hackathon at EPFL Apr. 2024 a two-day hackathon on improving LLM architectures like GPT 2 and Llama 2 held at EPFL. ◇ Bronze Medal Sept. 2017 in the National Olympiad in Informatics. ◇ Ranked Second in !Optimizer Contest Sept. 2021 a nation-wide scientific contest in optimization
LEADERSHIP AND SERVICE	◇ Head of Cultural Student Association, EPFL Aug. 2023 Led cultural and scientific events with 250+ participants. ◇ Head of Student Scientific Association June 2019 Led student scientific activities in the Department of Mathematical Sciences. ◇ Co-organizer, Mathematical Sciences Summer School Aug. 2019 Organized a summer school for 80 high-school students covering mathematics and computer science topics.
SKILLS	◇ Vision and Gen-AI: Video Generation, World Models, Diffusion Models, 4D Gaussian Splats ◇ Deep Learning: PyTorch, NumPy, JAX (familiar), Slurm, Distributed Training ◇ Programming Languages: Python, Java (familiar), C++ (familiar) ◇ Tools: Git, Docker, WandB
LANGUAGES	◇ English (fluent, TOEFL: Total 110) ◇ French (B1, intermediate)